TDP005 Projekt: Objektorienterat system

Space survival

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1 Space survival

1.1 HowTo

1.1.1 Run game

make ./prog

1.1.2 Documentation

```
See files in /docs
```

2 Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

std::enable_shared_from_this

- **State** 7
  - **Exit_State** 4
  - **Game_State** 5
  - **Menu_State** 5
- **Game_Object** 4
  - **Enemy** 3
    - **Asteroid** 3
    - **Bullet** 3
    - **Missile** 6
    - **Ufo** 7
  - **Player** 6
  - **Projectile** 6
- **Texture_Manager** 7
- **World** 8

3 Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Asteroid 3

Bullet 3

Enemy 3

Exit_State 4

Game_Object 4

Game_State 5

Menu_State 5

Missile 6

Player 6

Projectile 6

State 7

Texture_Manager 7

Ufo 7

World 8

4 Class Documentation

4.1 Asteroid Class Reference

Inheritance diagram for Asteroid:

4.2 Bullet Class Reference

Inheritance diagram for Bullet:

Collaboration diagram for Bullet:

Public Member Functions

- **Bullet** (const sf::Vector2f &pos, const float speed, const std::string &texture, const sf::Vector2f &player)
- bool **update** (sf::Time delta, World &World) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.3 Enemy Class Reference

Inheritance diagram for Enemy:

Collaboration diagram for Enemy:

Public Member Functions

- **Enemy** (const std::string &texture, const float s)

Protected Member Functions

- void **out_of_bounds** ()

Protected Attributes

- const float **speed**
- const double **random**

Additional Inherited Members

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.4 Exit_State Class Reference

Inheritance diagram for Exit_State:

Collaboration diagram for Exit_State:

Public Member Functions

- std::shared_ptr< State > **update** (sf::Time)
- void **render** (sf::RenderWindow &)

Additional Inherited Members

The documentation for this class was generated from the following file:

- state.h

4.5 Game_Object Class Reference

Inheritance diagram for Game_Object:

Public Member Functions

- **Game_Object** (const std::string &texture)
- virtual bool **update** (sf::Time time, World &world)=0
- void **render** (sf::RenderWindow &window)

Public Attributes

- sf::RectangleShape **shape**
- sf::Vector2f **dir**

The documentation for this class was generated from the following files:

- game_object.h
- game_object.cc

4.6 Game_State Class Reference

Inheritance diagram for Game_State:

Collaboration diagram for Game_State:

Public Member Functions

- std::shared_ptr< State > **update** (sf::Time delta) override
- void **render** (sf::RenderWindow &to) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- game_state.h
- game_state.cc

4.7 Menu_State Class Reference

Inheritance diagram for Menu_State:

Collaboration diagram for Menu_State:

Public Member Functions

- **Menu_State** (std::shared_ptr< State > resume=nullptr, int score=0)
- void **on_key_press** (sf::Keyboard::Key key) override
- std::shared_ptr< State > **update** (sf::Time time) override
- void **render** (sf::RenderWindow &draw) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- menu_state.h
- menu_state.cc

4.8 Missile Class Reference

Inheritance diagram for Missile:

Collaboration diagram for Missile:

Public Member Functions

- **Missile** (const sf::Vector2f &pos, const float speed, const std::string &texture)
- bool **update** (sf::Time delta, World &World) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.9 Player Class Reference

Inheritance diagram for Player:

Collaboration diagram for Player:

Public Member Functions

- **Player** (const sf::Vector2f &pos)
- bool **update** (sf::Time delta, World &world) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- player.h
- player.cc

4.10 Projectile Class Reference

Inheritance diagram for Projectile:

Collaboration diagram for Projectile:

Public Member Functions

- **Projectile** (const sf::Vector2f &pos, const float r)
- bool **update** (sf::Time delta, World &world) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- projectile.h
- projectile.cc

4.11 State Class Reference

Inheritance diagram for State:

Collaboration diagram for State:

Public Member Functions

- virtual void **on_key_press** (sf::Keyboard::Key key)
- virtual std::shared_ptr< State > **update** (sf::Time time)=0
- virtual void **render** (sf::RenderWindow &to)=0

Static Public Member Functions

- static void **run** (sf::RenderWindow &window, std::shared_ptr< State > state)

The documentation for this class was generated from the following files:

- state.h
- state.cc

4.12 Texture_Manager Class Reference

Static Public Member Functions

- static sf::Texture * **get** (const std::string &name)

The documentation for this class was generated from the following files:

- texture_manager.h
- texture_manager.cc

4.13 Ufo Class Reference

Inheritance diagram for Ufo:

Collaboration diagram for Ufo:

Public Member Functions

- **Ufo** (const sf::Vector2f &pos, const float speed, const std::string &texture)
- bool **update** (sf::Time delta, World &World) override

Additional Inherited Members

The documentation for this class was generated from the following files:

- enemy.h
- enemy.cc

4.14 World Class Reference

Public Member Functions

- void **update** (sf::Time delta)
- void **render** (sf::RenderWindow &to)
- void **add** (shared_ptr< Game_Object > object)
- std::vector< shared_ptr< Game_Object > > **collides_with** (Game_Object &me) const
- bool **is_dead** () const
- sf::Vector2f **player_pos** () const
- int **get_score** ()
- int **get_maxEnemies** ()
- int **get_ufo_num** ()
- void **add_score** (int s)
- void **add_maxEnemies** (int m)
- void **add_ufo_num** (int u)

The documentation for this class was generated from the following files:

- world.h
- world.cc

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